

ResStock demo, market results

Local energy sharing AMM on measured building data and real day-ahead prices

Community	10 prosumers (5 with rooftop PV, 5 without), single Massachusetts county, ResStock amy2018 release 2
Period	4 trading days, 2018-07-01 to 2018-07-04, 96 sessions of 15 minutes per day
Prices	ISO New England day-ahead LMP, zone WCMA, retail = wholesale + 0.12 EUR/kWh delivery adder
Status quo	every kWh traded with the grid alone: exports paid low (wholesale), imports paid high (retail)

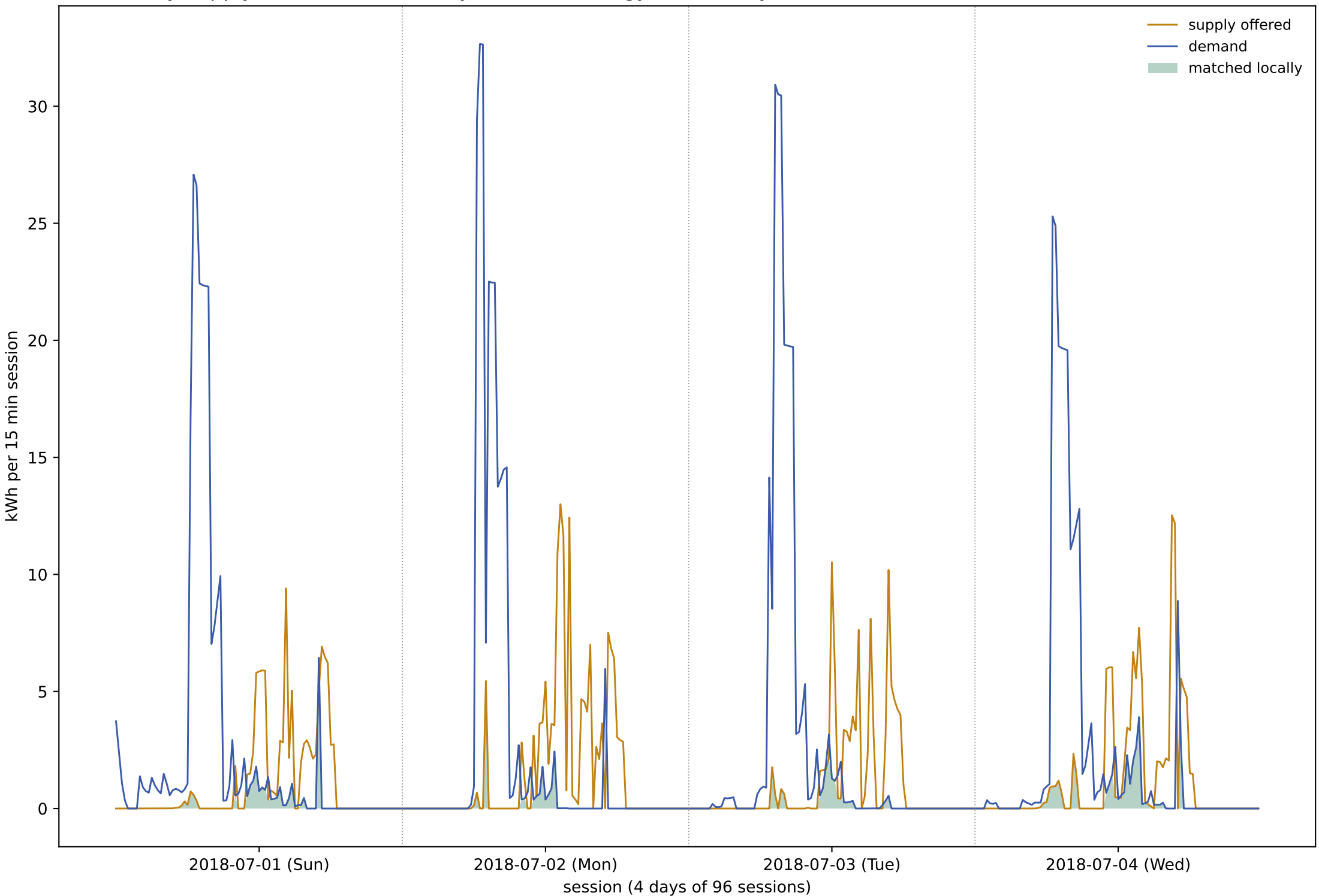
Headline results

- Community gain over 4 days: 9.49 EUR (2.37 EUR/day), entirely the delivery spread earned on locally matched energy
- Locally matched energy: 79.1 kWh of 955.7 kWh demanded, 8.3 % of demand served locally (17.1 % of the surplus absorbed locally)
- Sellers received on average 7.72 c/kWh against 6.69 wholesale; buyers paid 15.00 c/kWh against 15.49 retail
- Spread high - low = the 12 c/kWh delivery adder in every session, so the gain is 0.12 EUR per matched kWh and matching volume is the only lever

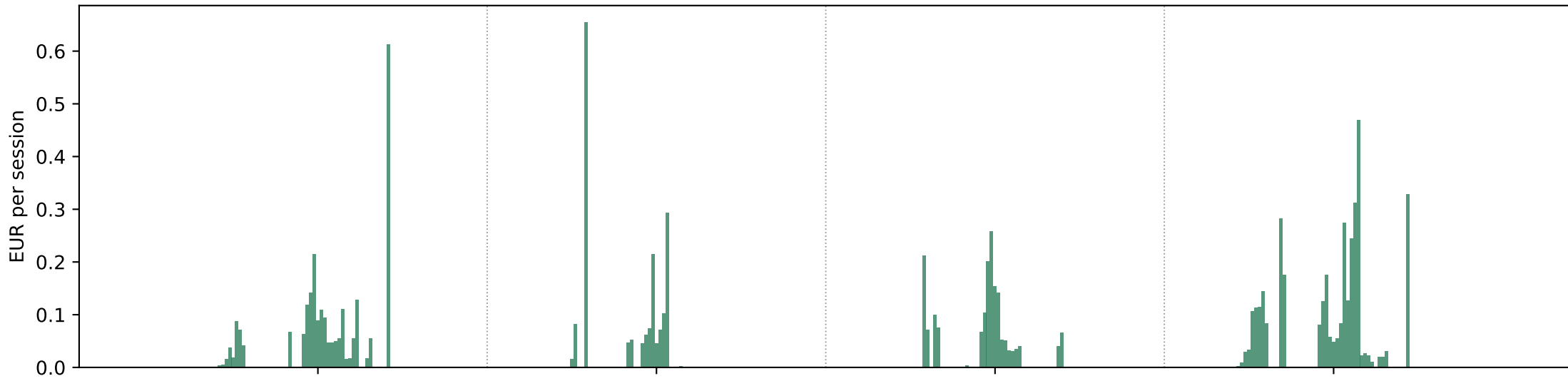
Daily market summary

	offered kWh	demanded kWh	matched kWh	covered %	absorbed %	gain EUR
2018-07-01 (Sun)	102.50	244.88	19.85	8.11	19.36	2.38
2018-07-02 (Mon)	143.91	249.02	14.67	5.89	10.19	1.76
2018-07-03 (Tue)	99.10	233.72	14.43	6.17	14.56	1.73
2018-07-04 (Wed)	116.12	228.10	30.17	13.23	25.98	3.62

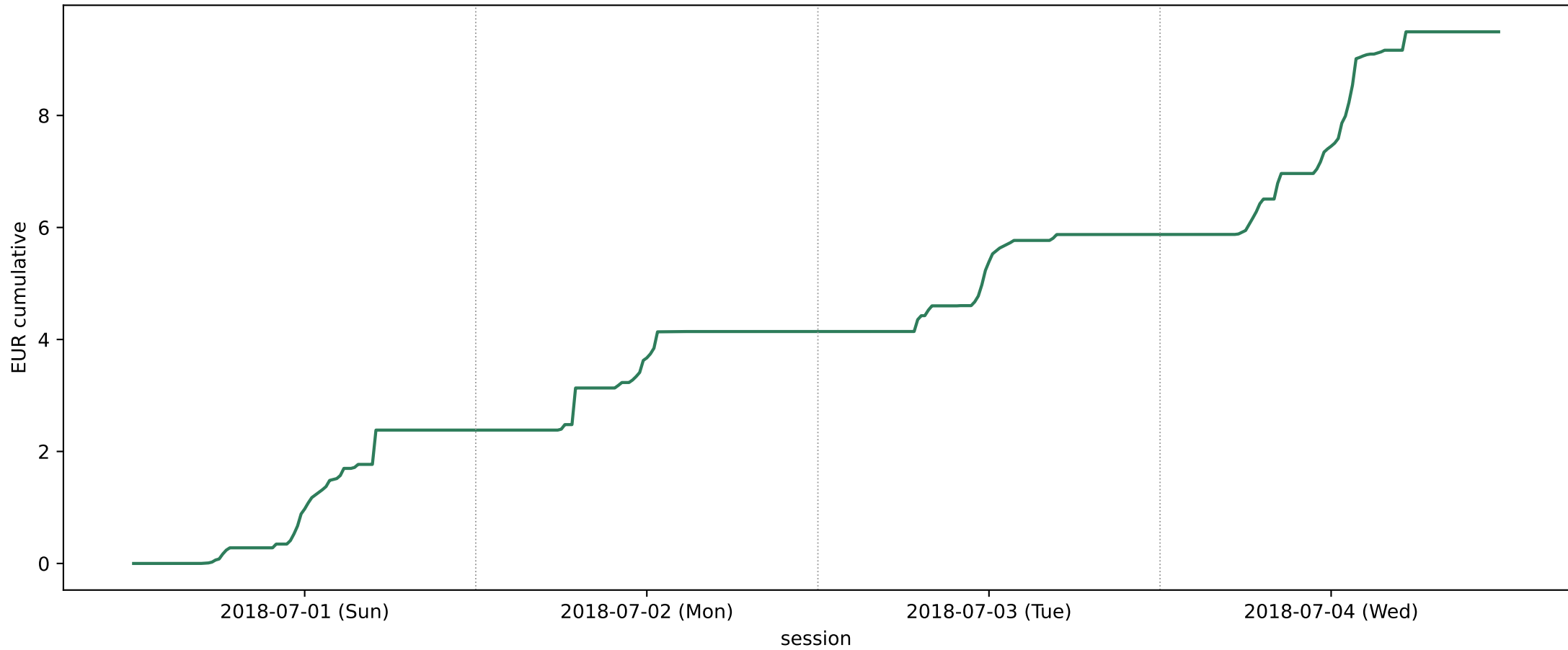
Community supply, demand and locally matched energy, session by session



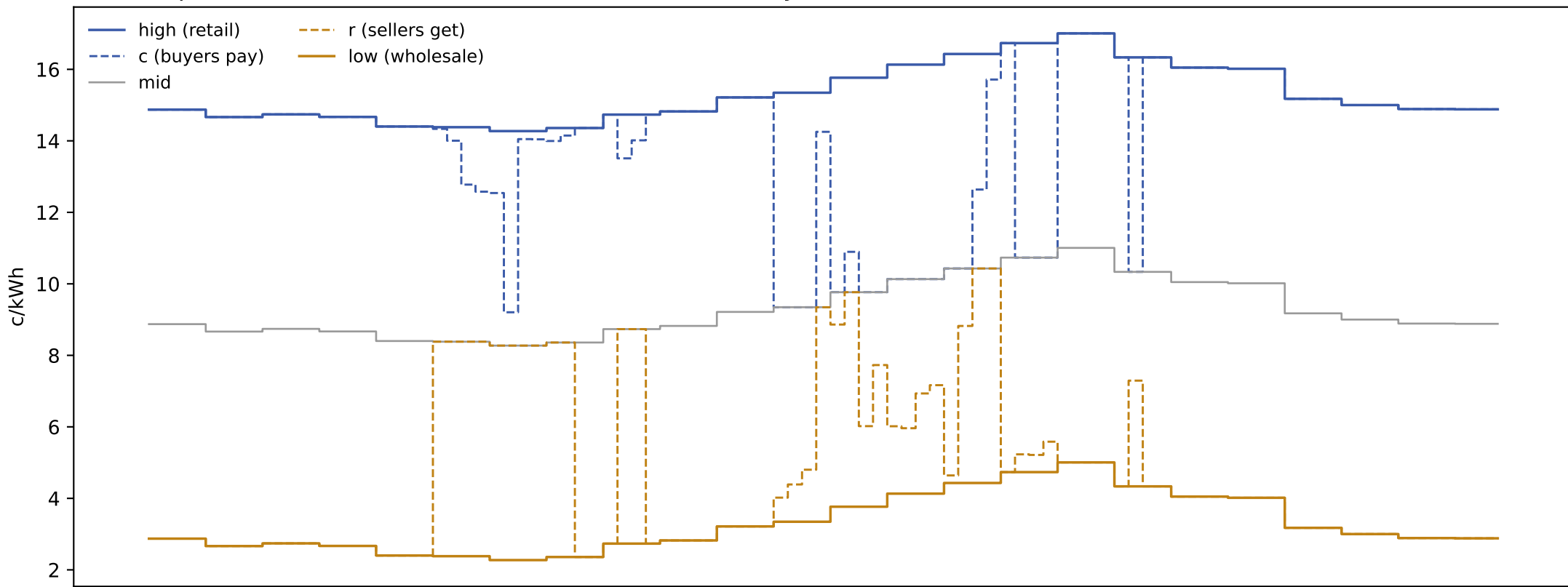
Community gain per session versus the status quo



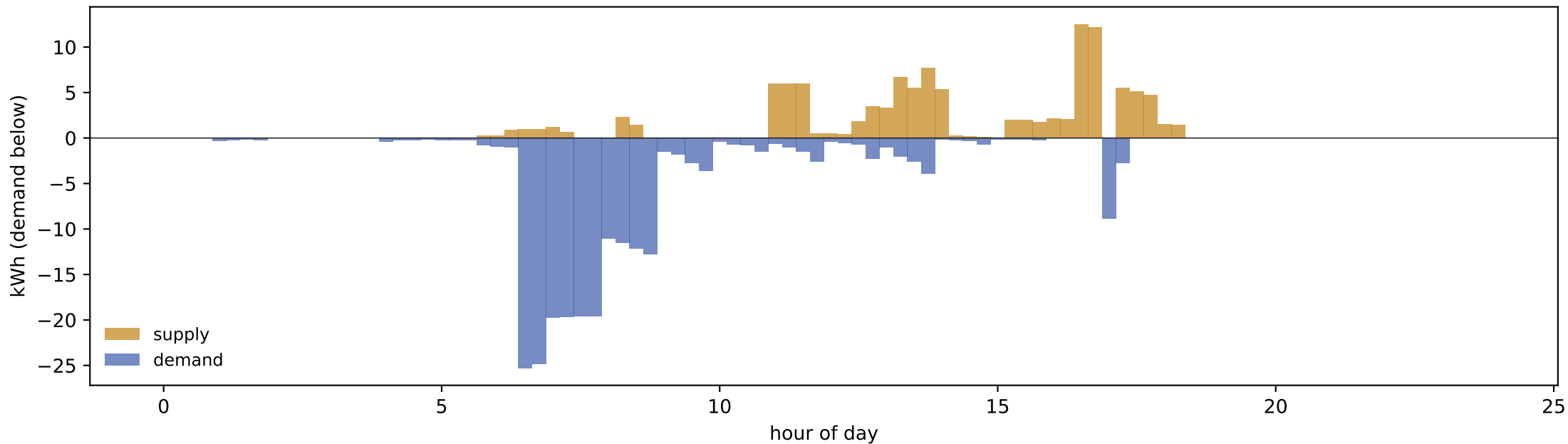
Cumulative community gain across the run



Session prices on 2018-07-04 (Wed), the most active day



The side in excess pushes its price toward the grid price of that side

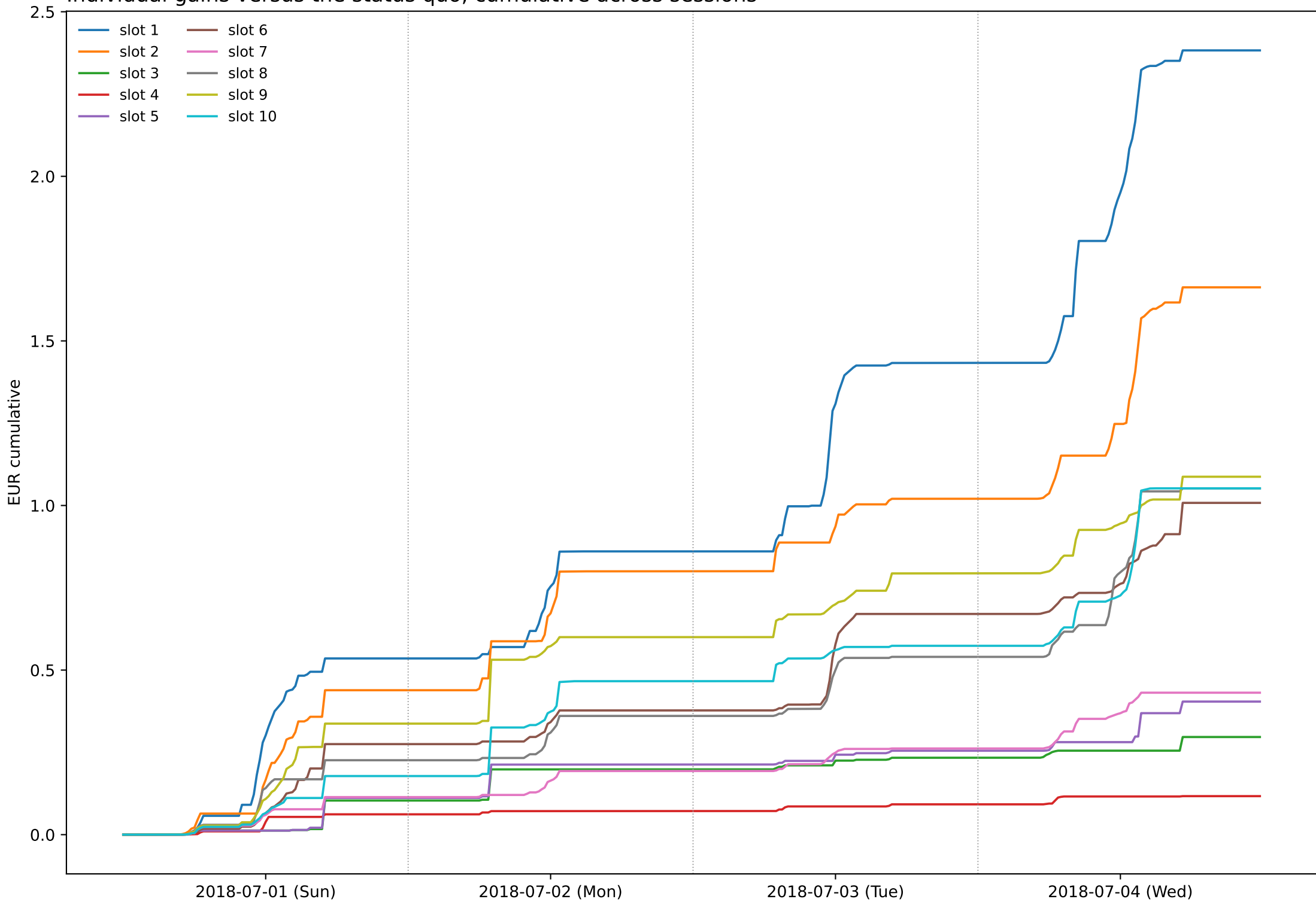


Individual results over the whole run

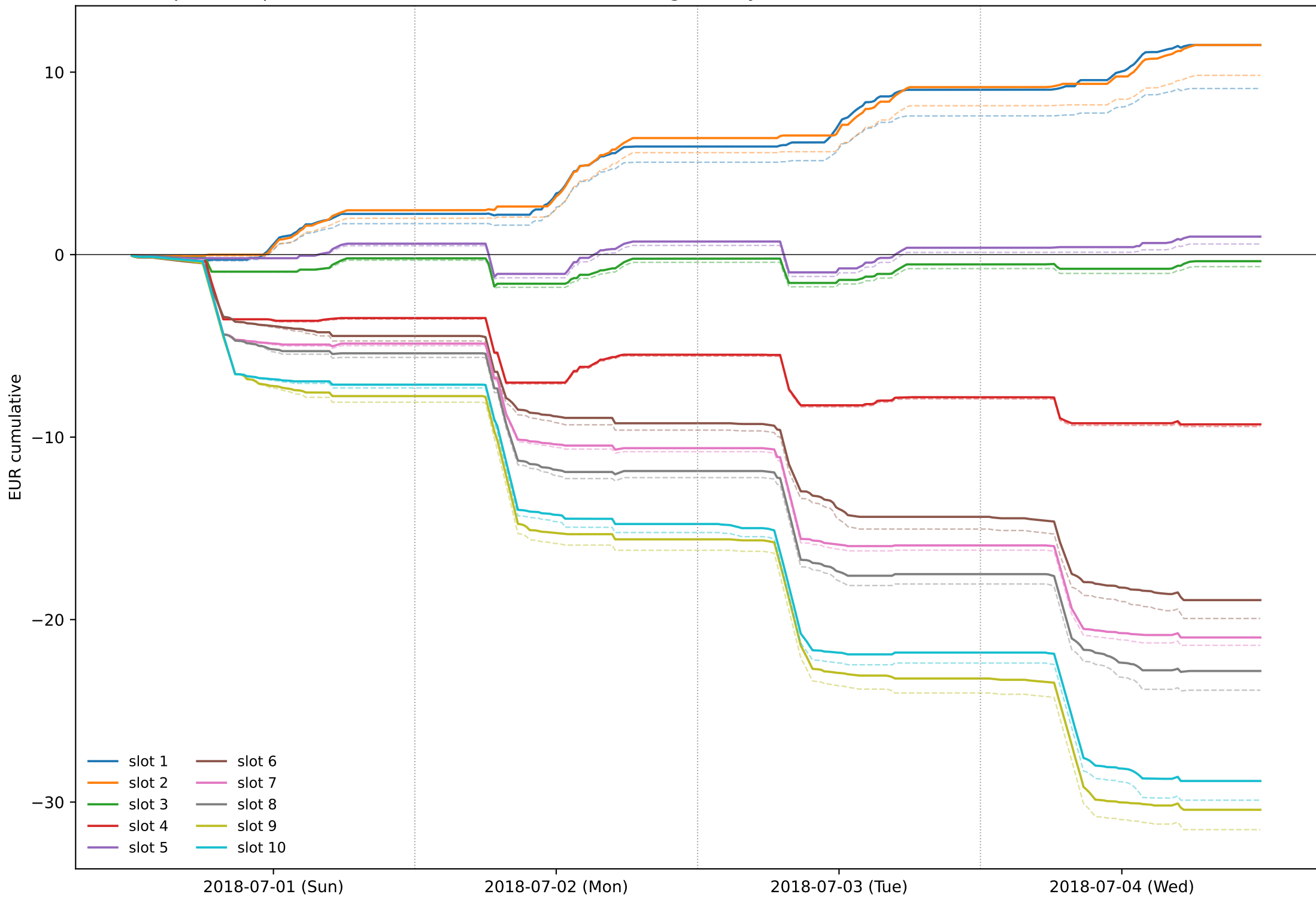
Cash flows in EUR, negative = net payment. Gain = with AMM - grid only; it is non negative for every member in every session.

	sold kWh	bought kWh	grid only EUR	with AMM EUR	gain EUR	gain c/kWh traded	share of gain %
slot 1	153.46	3.69	9.10	11.48	2.38	1.516	25.1
slot 2	153.24	0.91	9.82	11.49	1.66	1.079	17.5
slot 3	48.77	26.86	-0.66	-0.36	0.30	0.392	3.1
slot 4	26.10	76.56	-9.42	-9.30	0.12	0.114	1.2
slot 5	60.19	24.15	0.58	0.99	0.40	0.479	4.3
slot 6	1.82	127.66	-19.94	-18.93	1.01	0.778	10.6
slot 7	5.99	141.55	-21.41	-20.98	0.43	0.292	4.5
slot 8	6.53	156.29	-23.87	-22.82	1.05	0.646	11.1
slot 9	2.24	203.52	-31.51	-30.42	1.09	0.528	11.5
slot 10	3.29	194.52	-29.90	-28.84	1.05	0.532	11.1

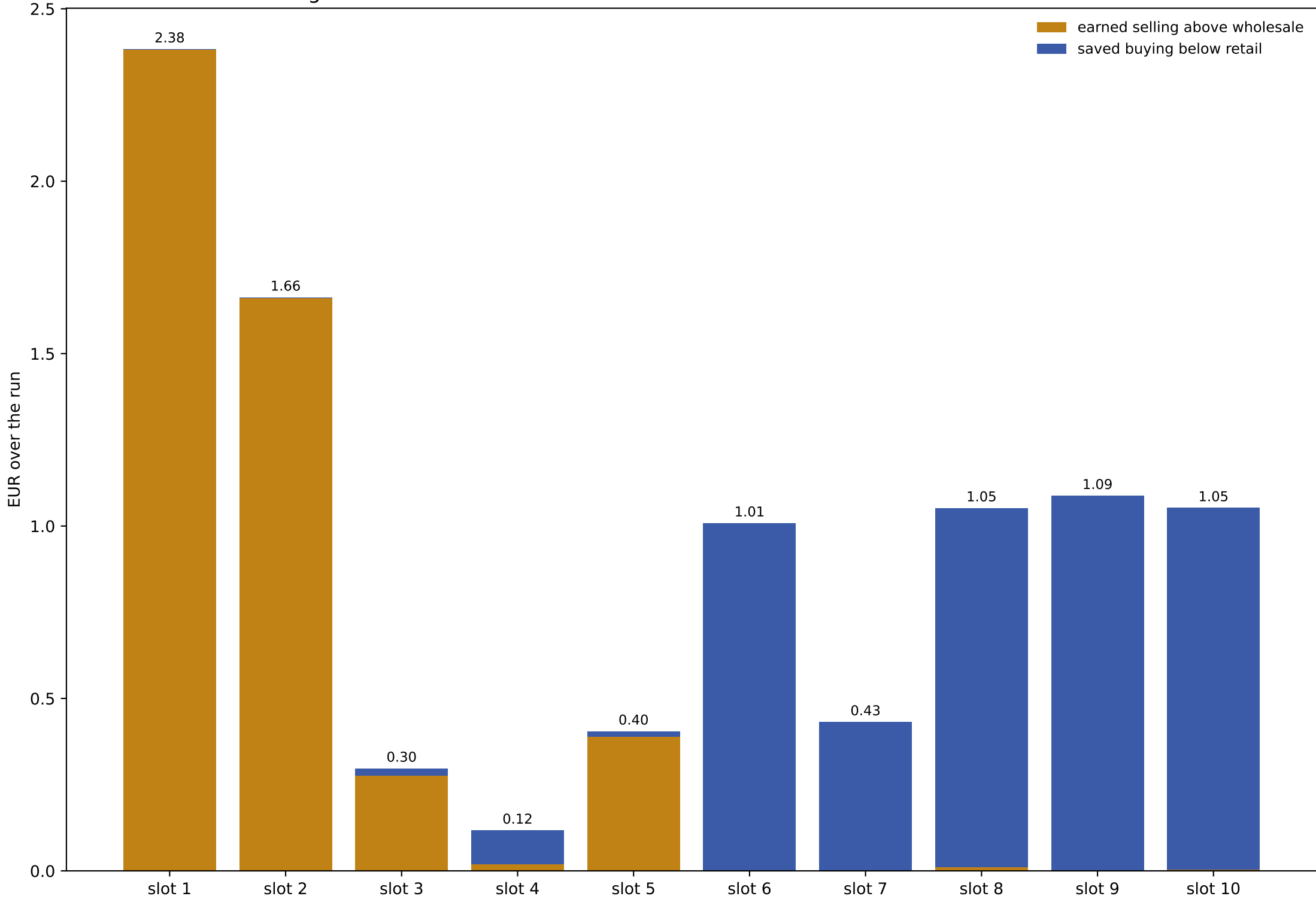
Individual gains versus the status quo, cumulative across sessions



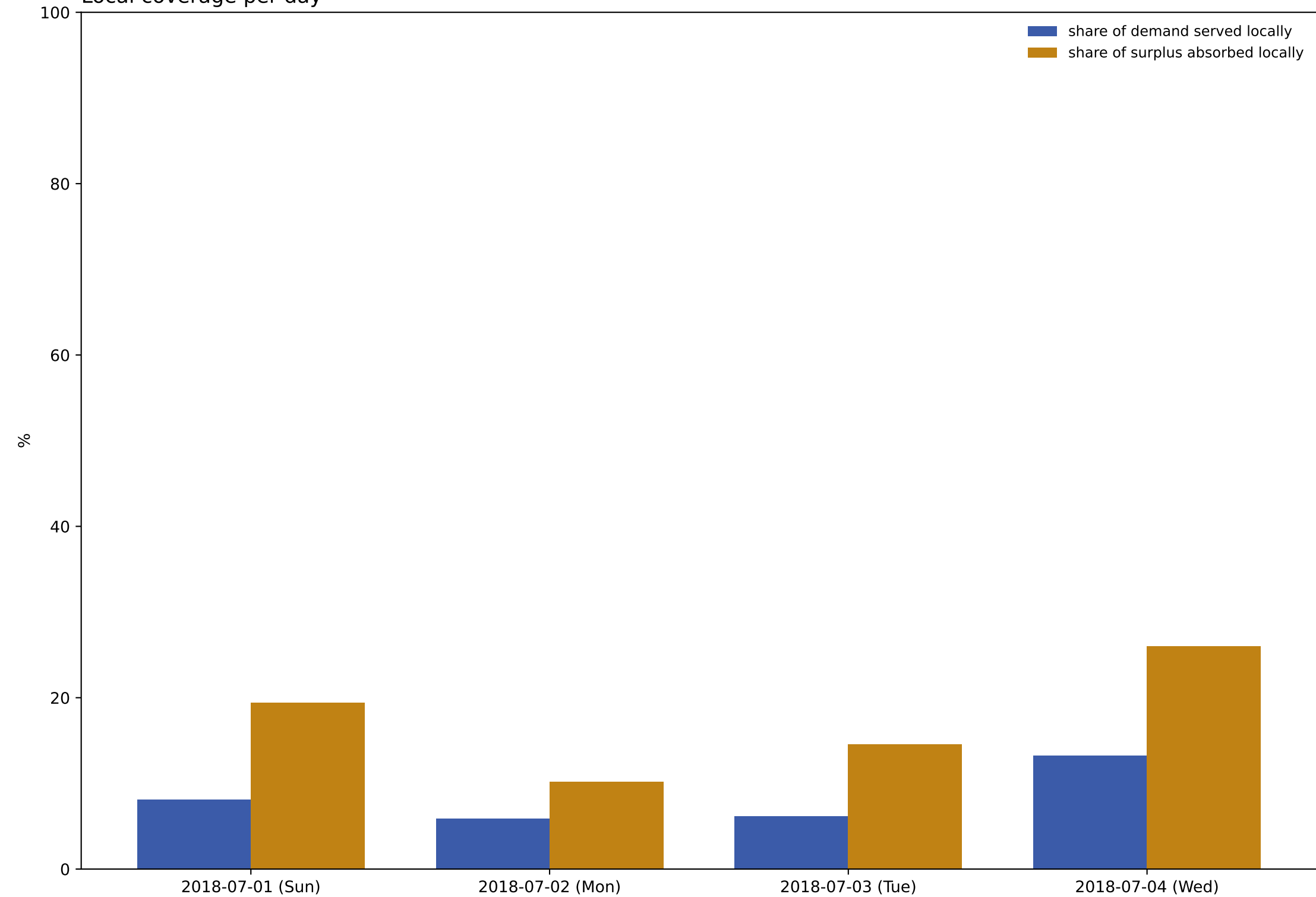
Net cash position per member, with the AMM (solid) and grid only (dashed)



Where each member's gain comes from



Local coverage per day



Average gain per household by community size, over random sub-communities (net positions held fixed, as price-taking makes them

